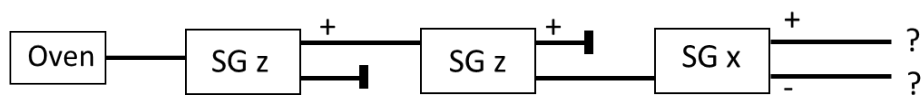


1. Give an example to show the inverse feature of linear vector space, obeyed by 2X2 matrices. **(2)**
2. In the vector space of real polynomials $p(x)$ (real variable x with real coefficients), consider operators T and S , denoting differentiation and multiplication with x respectively. Verify that T and S is a linear operator. Check its commutator. **(4)**
3. Given $R(\pi/2) |1\rangle = |1\rangle$; $R(\pi/2) |2\rangle = -|3\rangle$; $R(\pi/2) |3\rangle = -|2\rangle$. Check whether R is unitary. **(6)**
4. Write down the completeness relation for spin state of an electron. **(2)**
5. Commutator of 2 Hermitian operator is: **(2)**
a) Hermitian b) Anti-Hermitian c) Unitary d) Zero
6. Check if it is possible to have simultaneous eigenket of S and T if S and T anti-commute. **(4)**
7. Using Gram-Schmidt theorem convert given basis into orthonormal one. **(6)**
 $|1\rangle = \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix}$; $|2\rangle = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$; $|3\rangle = \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix}$
8. Find eigenvalue and eigenvector of (a) projection operator (b) identity operator **(4+2)**
9. Check whether the momentum operator of a free particle is a conserved operator. **(3)**
10. For the 2 configurations of sequential Stern-Gerlach experiment given below, the 2 components of spin at the output in both cases is: (SG z stands for in-homogenous magnetic field in z direction etc.) **(4)**

(a)



(b)

